

HARD LINK AND SOFT LINK IN UNIX

Complete Technical Reference Guide

Fundamental Concept

Before understanding links, remember how UNIX maps files to physical storage:

Filename → Inode → Actual Data

Example: notes.txt (Filename) → **Inode 125** (Metadata) → **"Hello UNIX"** (Data).

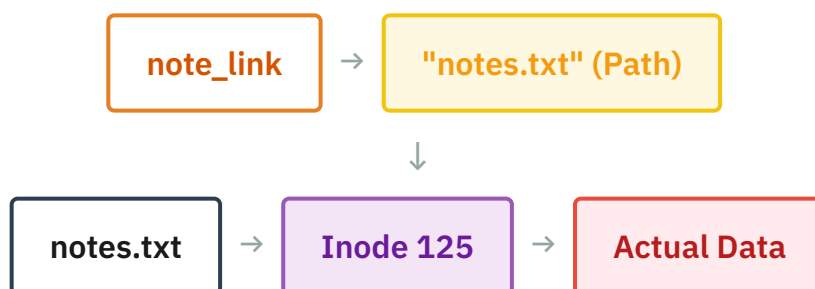
The filename is just a label in a directory. The Inode stores file info (permissions, owner, timestamps), and the Inode points to the actual data blocks.

1. Soft Link (Symbolic Link)

A soft link is a special type of file that contains a reference to another file or directory in the form of an absolute or relative path. It behaves like a shortcut.

```
$ ln -s notes.txt note_link
```

Internal Working (Soft Link)



When you access `note_link`, UNIX reads the stored path, finds `notes.txt`, and follows it to the data. **If the original is deleted:** The path becomes invalid, resulting in a **broken link**.

Summary:

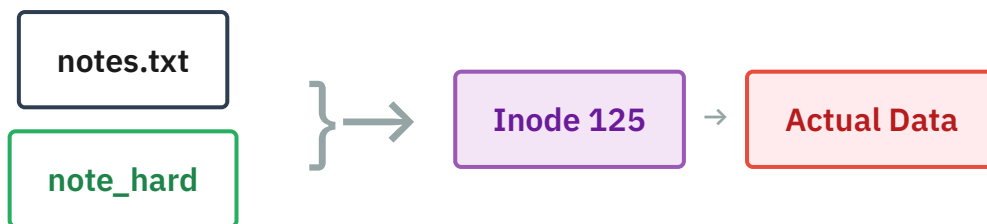
- Stores pathname of another file.
- Has its own unique inode number.
- Can link files and directories.
- Can cross file systems.

2. Hard Link

A hard link is another filename that points directly to the **same inode** as the original file. Both filenames are equal references to the same content.

```
$ ln notes.txt note_hard
```

Internal Working (Hard Link)



Both names access the same Inode (125). **If `notes.txt` is deleted**, the inode is not removed because `note_hard` still points to it. The data is only deleted when 100% of hard links are gone.

Summary:

- Points directly to the same Inode.
- Both filenames are identical references.
- Survives deletion of the original filename.

- Cannot link directories or cross file systems.

Difference Table

Feature	Hard Link	Soft Link
Points To	Inode	Path/Filename
Uses Same Inode	Yes	No (Has its own)
Cross File Systems	No	Yes
Links Directories	Generally No	Yes
Original Deleted	Link Still Works	Link Becomes Broken
Memory Usage	Zero extra storage	Stores path string

MEMORY TRICK:

- **Hard Link** = Same Inode (A clone name)
- **Soft Link** = Stores Path (A signpost)

If Original is Deleted: Hard link stays alive; Soft link dies.